Table S1. Population parameters from literature for the Scalloped Hammerhead Shark (Sphyrna lewini).

Parameter	Key	Sex	Value	Derived parameter	Source	Notes
Age at maturity	amat	Female	13	Natural mortality (M), Time at stage (Ti)	Anislado-Tolentino et al 2008	Mexican Pacific
		Male	8		Anislado-Tolentino et al 2008	Mexican Pacific
		Female	13		Drew et al 2014	Indonecia
		Male	8		Drew et al 2014	Indonecia
		Female	160 cm tl		Alejo-Plata et al 2018	Oaxaca, Mexico
		Male	155 cm tl		Alejo-Plata et al 2018	Oaxaca, Mexico
		Female	219.4 cm tl		Estupiñan-Montaño 2021	Ecuador. 2003-2009 data
		Male	178.1 cm tl		Estupiñan-Montaño 2021	Ecuador. 2003-2009 data
Litter size	f	Female	16-22	Fecundity at stage (fi)	Estupiñan-Montaño 2021	Ecuador. 2003-2009 data
		Female	25.8		Huy and Tsai, 2023. Shr 2020	Taiwan
		Female	14-41		White et al 2008	Indonesia
		Female	15-42		Alejo-Plata et al 2018	Oaxaca, Mexico

Growth rate	g	Female	0.1	Natural mortality (M), Longevity (amax)	Anislado-Tolentino et al 2008	Mexican Pacific
		Male	0.12		Anislado-Tolentino et al 2008	Mexican Pacific
		Combined	0.159		Drew et al 2014	Indonecia
		Female	0.161		Drew et al 2014	Indonecia
		Male	0.155		Drew et al 2014	Indonecia
		Female	0.142		Huy and Tsai, 2023. Shr 2020	Taiwan
		Male	0.165		Huy and Tsai, 2023. Shr 2020	Taiwan
		Combined	0.1199		Liu et al. 2021	Gulf of Mexico. Simmulated
L infinity	Linf	Female	376	Natural mortality (M), Longevity (amax)	Anislado-Tolentino et al 2008	Mexican Pacific
		Male	364		Anislado-Tolentino et al 2008	Mexican Pacific
		Female	331		Alejo-Plata et al 2018	Oaxaca, Mexico
		Male	308		Alejo-Plata et al 2018	Oaxaca, Mexico
		Combined	289.6		Drew et al 2014	Indonecia
		Female	289.6		Drew et al 2014	Indonecia
		Male	259.8		Drew et al 2014	Indonecia
		Female	367.9		Huy and Tsai, 2023. Shr 2020	Taiwan

		Male	317.7		Huy and Tsai, 2023. Shr 2020	Taiwan
		Combined	285.51		Liu et al. 2021	Gulf of Mexico. Simmulated
Length at birth	L0	Combined	54.2	Longevity (amax)	Drew et al 2014	Indonecia
		Female	53.2		Drew et al 2014	Indonecia
		Male	56.8		Drew et al 2014	Indonecia
		Combined	47-55		Estupiñan-Montaño 2021	Ecuador. 2003-2009 data
		Female	48.5		Huy and Tsai, 2023. Shr 2020	Taiwan
		Male	48.5		Huy and Tsai, 2023. Shr 2020	Taiwan
		Combined	41.1		Liu et al. 2021	Gulf of Mexico. Simmulated
		Combined	44-54		Alejo-Plata et al 2018	Oaxaca, Mexico
Maximum age	amax	Female	12.5	Natural mortality (M), Time at stage (Ti)	Anislado-Tolentino et al 2008	Mexican Pacific
		Male	11		Anislado-Tolentino et al 2008	Mexican Pacific
		Combined	35		Drew et al 2014	Indonecia
		Female	24.41		Huy and Tsai, 2023. Shr 2020	Taiwan
		Male	21.1		Huy and Tsai, 2023. Shr 2020	Taiwan

Reproductive cycle	Female	2	Huy and Tsai, 2023. Shr 2020	Taiwan

Table S2. Scenario shows the management scenario. Mean and sd lambda show the mean and standard deviation of population growth (lambda) across simulations for each scenario. Lambda >1 shows the percentage of simulations where population growth was larger than 1. "Current" shows the values under the estimated current conditions for the scalloped hammerhead population in the ETP. 2x and 3x juvenile fishing multiply juvenile and neonate fishing mortality from table 2 times two and three times respectively. The Extreme Elastic scenario is equal to 3xjuvenile fishing but includes a 75% reduction of adult and subadult fishing mortality. Neonates protected reduces neonate fishing mortality by 90%. Juveniles protected reduces neonate and juvenile fishing mortality by 75%. The 25% and 50% reduction scenario assumes fishing mortality is reduced for all life stages. The adults protected scenario assumes a 75% reduction in fishing mortality for subadults and adults. The no fishing scenario assumes that fishing mortality is eliminated for the population.

Scenario	mean_lambda	sd_lambda	lambda > 1
Current	0.860	0.054	0.94%
2xJuvenile_fishing	0.808	0.047	0.03%
3xJuvenile_fishing	0.771	0.042	0.01%
Extreme Elastic	0.915	0.034	0.85%
Neonates_protected	0.896	0.060	5.05%
Juveniles_protected	0.910	0.061	7.71%
25%_reduction	0.925	0.054	8.64%
50%_reduction	0.996	0.055	45.35%
Adults_protected	1.020	0.050	63.82%
No_fishing	1.162	0.062	99.93%

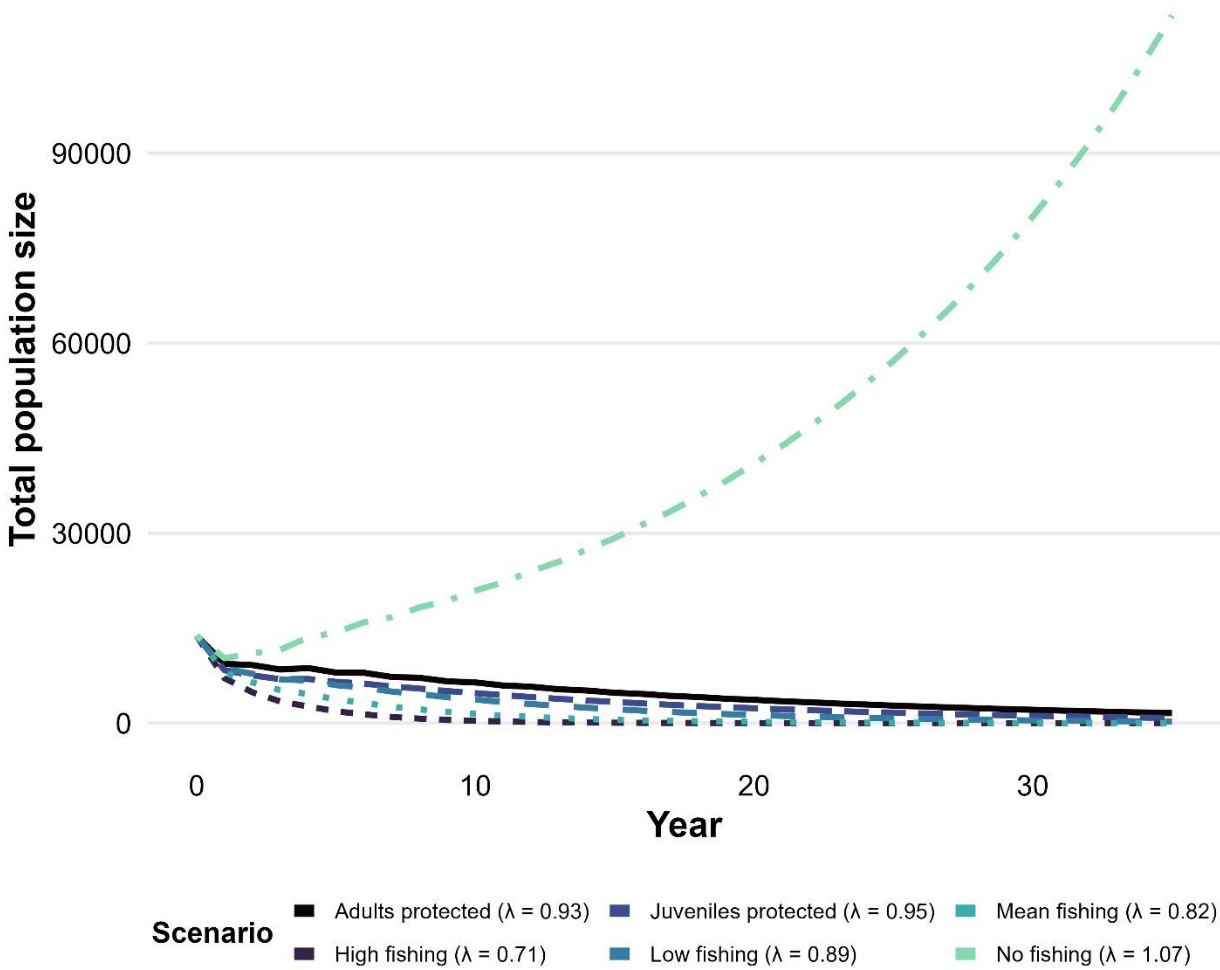

Figure S1. Projected population growth of a simulated Sphyrna lewini population using fixed values as entries of the population model. Scenarios show variation in the Fishing Mortality parameter. Lambda (λ) values show the rate of population increase (if λ >1) or decrease (if λ <1)

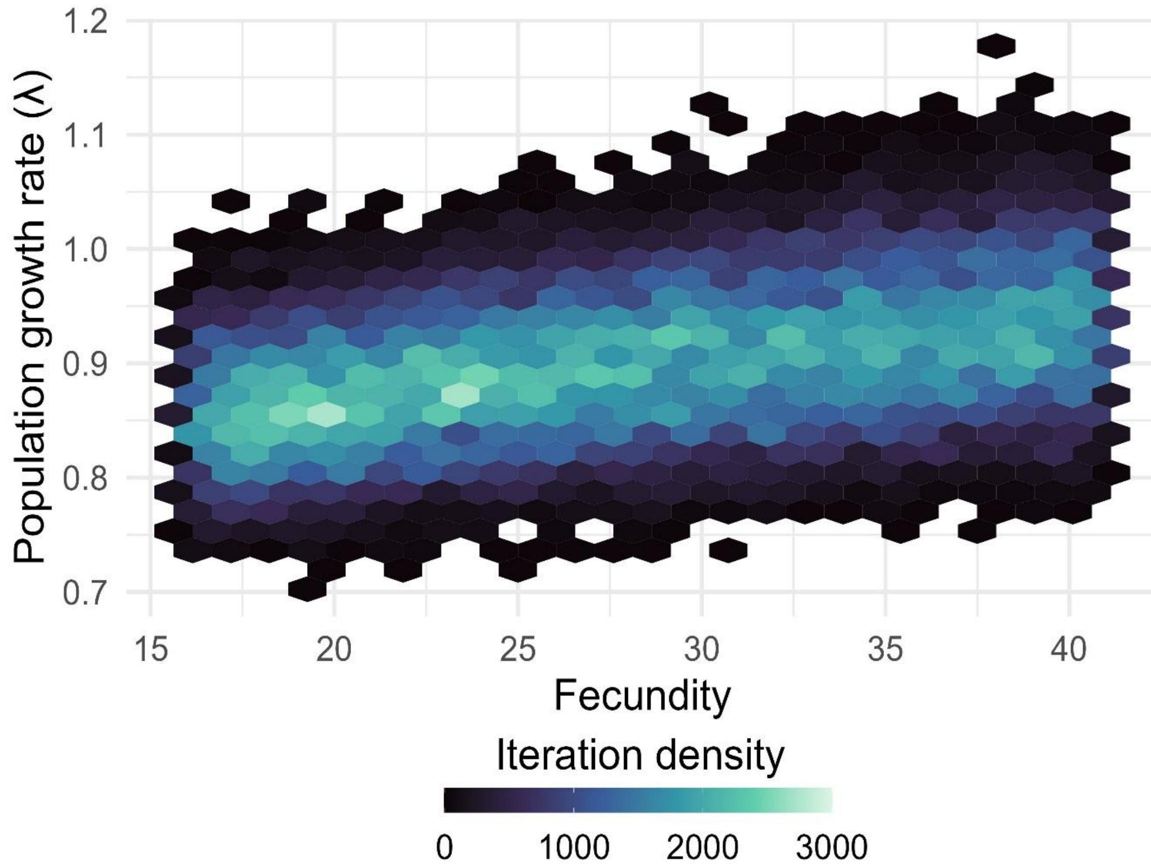

Figure S2. Variation in population growth estimates as a function of fecundity across all simulated scalloped hammerhead virtual populations (N = 18,000). The y-axis shows population growth (λ), and the x-axis shows fecundity values. Color variation within each hexagon represents the density of iterations corresponding to each λ–fecundity combination, with lighter colors indicating higher densities.